
Client Decision Report

Janglish: LID-Guided Transcription of Japanese–English Code-Switched Speech

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CS 6351 · AI & ML Project 2

Decision in one sentence. Keep the system as human-reviewed transcription assistance for research and personal use; extend Japanese-dominant transcription to a small monitored beta pilot built on the synthetic-only checkpoint, and hold English-dominant support back for further research.

Executive Recommendation

Recommendation: keep the system as human-reviewed transcription assistance for research and personal use, and run a small monitored beta pilot for Japanese-dominant audio only. The system may produce a first-draft transcript of Japanese–English code-switched speech, but a bilingual reviewer must inspect and correct every transcript before it is used or shared. English-dominant sentences with embedded Japanese are not ready: on that direction the pipeline can perform worse than plain Whisper and requires further research before any user-facing claim.

This recommendation balances a meaningful technical success against unresolved reliability and rights concerns. The fine-tuned frame-level language-identification (LID) model (wav2vec2-xls-r-300m with a 2-class frame head) reached macro-F1 0.9760 and frame accuracy 0.9768 on speaker-disjoint held-out data, and the LID-guided splice pipeline cut word error rate on the hardest real speaker from 0.5516 to 0.2018. However, the held-out set contains no real human voices, English-dominant audio showed regressions, the splice gates are uncalibrated, spontaneous conversation is untested, and the training data embeds licensing and consent constraints (XTTS v2's non-commercial license; contributor recordings collected without specific written terms).

The client's next action should be a monitored pilot, not a public release. The pilot should be restricted to Japanese-dominant audio, use the synthetic-only checkpoint lid-wav2vec2-f1-974 (trained while contributor audio was absent from the dataset, so no consent question attaches to it), log every correction, and build the missing real-voice evaluation evidence before any broader use.

Part A: Client Problem, Intended Use, And Decision Context

Client Problem And Stakeholders

Bilingual Japanese–English speakers rarely stay in one language. A speaker will say **娘はとても** independent **で、いつも自分で** choice **をしたがります。** — a single sentence that hops language mid-stream (“code-switching,” or informally “Janglish”). Whisper, the workhorse open transcription model, decodes each pass under one language assumption: in Japanese mode it renders embedded English as katakana; in English mode it can mangle Japanese into romaji or hallucinated English. Anyone who needs accurate transcripts of such speech — bilingual individuals and families, language learners, researchers — currently has no good option.

The primary stakeholder is the person relying on the transcript. Other affected stakeholders include the four contributors whose recorded voices are in the dataset, the roughly 165 Common Voice speakers whose reference clips seeded the synthetic voices, Coqui (whose XTTS v2 model generated the synthetic audio under a non-commercial license), and the project maintainer. A wrong transcript wastes correction effort or misleads its reader; releasing a model trained on contributor voices beyond what they agreed to would be a consent violation.

Target And Prediction-Time Inputs

The core model predicts one language label — English or Japanese — for every 20 ms frame of audio (50 decisions per second). At prediction time the system receives only 16 kHz mono audio; it must not use transcripts, filenames, or other metadata. Downstream, consecutive same-label frames become segments, and the “splice” strategy uses them to route Whisper passes: one pass in the dominant language, with English spans surgically replaced from a second English pass only when four safety gates agree. The end product is a draft transcript.

Intended, Unsupported, And Prohibited Uses

Permitted use:

- Research and personal use: first-draft transcripts of read or prepared Japanese–English code-switched speech, always reviewed by a bilingual human.

- A small, monitored beta pilot on Japanese-dominant audio under the conditions in Part C.
- Error analysis, model comparison, and further research on the code-switching problem.

Unsupported or prohibited use:

- Any commercial use while XTTS v2 synthetic audio (Coqui CPML, non-commercial) is in the training data.
- Public release of any checkpoint trained on contributor recordings; consent covered helping with a personal project, not redistribution.
- Autonomous or authoritative transcription — legal, medical, business records, or published subtitles without human review.
- Reliance on English-dominant transcription, which has measured regressions against baseline Whisper.
- Claims about spontaneous conversation; only read sentences have been evaluated.
- Displaying LID confidence or Whisper word probability as a calibrated probability.

Dataset And Remaining Assumptions

The manifest (`data/data-pairs.csv`) holds 31,634 labeled clips: 31,559 synthetic (2,515 pure English, 2,515 pure Japanese, and 26,529 code-switched clips bucketed by switch-island count `cs-1` through `cs-4`) and 75 real recordings from four speakers. Sentences were generated by Claude on AWS Bedrock (14,000+ lines) and filtered through five validators, including a ban on shared-phonology loanwords like “sushi” whose “switches” carry no acoustic evidence. Frame labels for synthetic audio are a free byproduct of a dual-render XTTS v2 synthesis pipeline plus forced alignment; failed alignments fall back to duration estimates and are flagged in sidecar files.

Remaining assumptions and limits: the corpus is 99.8% synthetic; reference voices come from Common Voice (low-quality clips were culled in June, leaving ~165); all audio is read speech, not conversation; labels within a 25 ms crossfade of each switch are approximations; contributor consent is informal; and the XTTS v2 license restricts commercial use of anything trained on its output.

Part B: Decision-Relevant Technical Evidence And Risk

Model Choice

The selected architecture is a fully fine-tuned `facebook/wav2vec2-xl-r-300m` — pretrained self-supervised on ~436k hours across 128 languages — with a frozen convolutional encoder and a 2-class frame-classification head at 20 ms resolution. Three checkpoints mark the project's progression.

Table 1: LID checkpoint comparison and decision meaning.

Model or checkpoint	Held-out macro-F1	Decision meaning
<code>wav2vec2-base</code>	not trained	Rejected early: English-only LibriSpeech pretraining offers weak Japanese representations; the multilingual XLS-R model costs little more compute.
<code>lid-wav2vec2-f1-823</code>	0.823	June 24 checkpoint: small early synthetic dataset, 40 epochs. Superseded by the data expansion, not by architecture changes.
<code>lid-wav2vec2-f1-974</code>	0.9743	July 5 checkpoint, trained on synthetic data only (contributor audio was absent; see appendix). The consent-clean candidate for the pilot.
<code>lid-wav2vec2-f1-98</code>	0.9760	July 7 checkpoint, full dataset including 75 real clips. Quality lead; research and personal use only.

The checkpoint comparison supports selection, not causal claims: runs differ in dataset size and epoch count. For the transcription backbone, Whisper base was chosen as a sub-300M-parameter default; a comparison against `whisper-large-v3` (and possibly `whisper-small`) is planned but has not been run, so it is unknown whether a larger Whisper alone would close part of the gap the pipeline provides. Among routing strategies, “splice” was

selected over “chop” and “dual” because its four gates fail toward baseline behavior: a rejected splice keeps the dominant-language words, so text is never dropped.

Metrics In The Client Context

Table 2: Frame-level metrics on speaker-disjoint held-out data.

Checkpoint	Accuracy	Macro-F1	F1 (EN)	F1 (JA)	Interpretation
f1-974	0.9751	0.9743	0.9790	0.9695	Synthetic-only training; pilot candidate.
f1-98	0.9768	0.9760	0.9804	0.9716	Full dataset; best current estimate.

Frame accuracy is the fraction of 20 ms frames labeled correctly; macro-F1 weights both languages equally. Splits are speaker-disjoint (grouped by speaker directory, seed 13), so no voice appears on both sides of the line — the most important honesty decision in the repository. But a replication of the split shows the 25 held-out test speakers are all synthetic: all four human contributors landed in the training partition. The frame metrics therefore measure generalization to unseen synthetic voices only, and say nothing measured about unseen real voices.

End-to-end evidence comes from the evaluation harness (faster-whisper base, int8, CPU; splice strategy; LID checkpoint f1-98), run August 1, 2026 on held-out synthetic test speakers and on the real recordings. References and transcripts are romanized to a shared alphabet before scoring, so a correct answer in a different script is not punished. Each speaker's sentences are roughly half English-primary and half Japanese-primary, so every row blends the strong direction with the weak one.

Table 3: End-to-end WER/CER, baseline Whisper vs. LID-guided pipeline (negative Δ is improvement). Speaker IDs abbreviated.

Evaluation slice	Files	Base WER	Base CER	Pipeline WER	Pipeline CER	Δ WER
cs-2/speaker_017	70	0.3247	0.1884	0.3218	0.1751	−0.0029
cs-4/speaker_017	9	0.5174	0.3035	0.6744	0.3687	+0.1570
cs-2/speaker_023	65	0.2710	0.1501	0.2990	0.1657	+0.0280
cs-4/speaker_023	9	0.4508	0.2054	0.4301	0.2027	−0.0207
cs-2/speaker_095	61	0.3743	0.2100	0.3326	0.1800	−0.0417
cs-4/speaker_095	13	0.5187	0.3106	0.4478	0.2500	−0.0709
manual/Johnny	20	0.2903	0.1353	0.2811	0.1267	−0.0092
manual/Ryan	20	0.1633	0.0781	0.1633	0.0789	+0.0000
manual/Sakamoto	20	0.5516	0.4007	0.2018	0.1098	−0.3498

Two caveats bound this table. The synthetic test speakers are truly held out from LID training; the three manual speakers are not (their 75 clips were in the training set of the evaluated checkpoint f1-98), so the manual rows may flatter generalization. And the pilot checkpoint f1-974 was not the one evaluated here; parity must be confirmed before launch.

Meaningful Success

The clearest win is the hardest real speaker: Sakamoto's baseline WER of 0.5516 — over half the words wrong, largely because Whisper flails on Japanese-dominant code-switched audio — dropped to 0.2018 (CER 0.4007 → 0.1098). This is exactly the user the system exists for. Six of nine evaluation slices improved, one tied, and the frame classifier's 0.976 macro-F1 on voices it never heard demonstrates that the synthetic-label factory — LLM sentences, dual-render TTS, forced alignment — produces supervision real enough to learn from. This evidence supports first-draft, human-reviewed use on Japanese-dominant audio. It does not support unreviewed use

anywhere: the same table contains a +0.157 WER regression.

Important Failures And Limitations

English-Dominant Regressions

On sentences that are primarily English with Japanese mixed in, the segmentation front-end tends to do worse than Whisper alone: it makes segmentation mistakes, and Whisper in English mode often (not always) recognizes embedded Japanese natively and emits correct kanji — so there is little headroom to win and real room to lose. One observed failure mode is segments far too short to be a plausible code switch. A third frame class for boundary/indecision regions might absorb this ambiguity without inventing false switches. The +0.1570 regression on cs-4/speaker_017 and the +0.0280 on cs-2/speaker_023 are the measured face of this weakness.

Boundary And Accent Sensitivity

The model does better when a slight pause precedes a code switch and struggles when the switch is smooth and natural — the register fluent bilinguals actually use. It also struggles with heavily Japanese-accented English words. No spontaneous conversation has been evaluated; every measured number comes from read sentences.

Synthetic Dominance And Held-Out Composition

With 99.8% of training audio synthesized by one TTS engine, the risk that the model partly recognizes XTTS texture rather than language cannot be fully excluded. The 75 real clips that guard against this are few, come from four speakers, and sit inside the training split — the held-out set contains no real voices at all. Sakamoto's result is encouraging real-world evidence, but it is one non-held-out speaker.

Uncalibrated Gates And Confidence

The four splice gates (span ≥ 0.3 s, mean LID confidence ≥ 0.85 , ≤ 3 words/s, mean word probability ≥ 0.5) were set by judgment, not tuned against data, and no calibration analysis exists for the LID softmax. Confidence values must not be shown to users or used for automatic acceptance until that evidence exists.

A Data-Handling Incident

In early July a synthetic-audio regeneration silently overwrote all 75 manual recordings with synthetic audio for the same sentences. The data was restored from version control on July 6 and a guard was added, but the incident shows the pipeline can modify source data without warning. It also produced an accidental asset: checkpoint f1-974, trained during the overwritten window, contains no contributor audio and is therefore free of the consent question (see appendix).

Responsible AI Risks That Affect The Decision

Table 4: Decision-relevant responsible AI risks.

Risk	Evidence status	Stakeholder consequence	Required boundary or action
Licensing and consent	XTTS v2 output is CPML (non-commercial); contributor consent was informal.	Commercial use or public release could violate the license or contributors' trust.	Research/personal use only; pilot uses the synthetic-only checkpoint; release requires a TTS swap and dropping or re-consenting contributor data.
Uneven direction reliability	Observed: English-dominant regressions up to +0.157 WER.	English-primary users can get worse output than plain Whisper.	Restrict pilot to Japanese-dominant audio; disclose the asymmetry.
Synthetic-doma in shift	99.8% synthetic corpus; no real voices in the held-out split.	Real-world quality may lag every reported frame metric.	Build a consented real-voice held-out set; keep human review.

Risk	Evidence status	Stakeholder consequence	Required boundary or action
Overconfidence and automation bias	Gates untuned; no calibration analysis.	Reviewers may accept plausible but wrong spliced words.	Hide confidence values; require review of every transcript.
Maintenance and reproducibility	Single maintainer; trained on one Mac in 6h51m; the Slurm cluster run failed silently.	The model cannot currently be retrained or verified elsewhere.	Document the environment; fix the cluster path before scaling.
Data integrity	Manual audio was silently overwritten once (restored).	Training or evaluation data can drift without notice.	Guard flag added; version manifests and snapshot before generation runs.

The intended use is low stakes compared with legal, medical, or employment decisions, which bounds likely harm. The realistic harms are wrong transcripts, wasted correction effort, misleading confidence, license violations, and quiet erosion of contributor trust.

Uncertainty That Limits The Recommendation

The following evidence is still missing:

- Any evaluation on spontaneous conversation rather than read sentences.
- A working English-dominant strategy (the third boundary/ambiguity frame class is untried).
- Calibration analysis for LID confidence and data-driven tuning of the four splice gates.
- A baseline comparison against whisper - large - v3 — whether a bigger model alone closes the gap.
- A held-out evaluation set of real voices under clear consent terms.
- End-to-end results for the pilot checkpoint f1-974 (Table 3 used f1-98).
- Latency, cost, and deployment measurements beyond CPU int8 evaluation runs.
- Written contributor terms and a rights review covering Common Voice attribution.

These gaps block public release and any English-dominant claim. They do not block a small Japanese-dominant, human-reviewed pilot, because every transcript remains editable and rejectable, the harm is bounded, and the pilot's purpose is to gather exactly the missing real-world evidence.

Part C: Responsible Use And Operational Plan

Permitted Workflow And Human Authority

The system may present a draft transcript alongside the source audio. The reviewer is the decision maker for every transcript and must be able to:

- listen to the original audio while reviewing;
- edit, accept, or discard the draft freely — including discarding the pipeline output in favor of a plain Whisper pass;
- flag failure cases: false code switches, missed smooth switches, accented-English errors, and implausibly short English spans;
- see that the transcript is a draft and that no confidence score is shown, because calibration has not been established.

Pilot conditions: Japanese-dominant audio only; LID checkpoint lid-wav2vec2-f1-974; beta testers receive written terms explaining what is collected and how corrections are used; every correction is logged with audio characteristics (pause vs. smooth switch, accent, read vs. spontaneous). The system must never autonomously publish, file, or certify a transcript.

Monitoring, Triggers, Responses, And Ownership

Table 5: Monitoring signals, triggers, responses, and ownership.

Signal	Trigger	Required response	Owner
WER delta vs. baseline on reviewed pilot batches	Pipeline worse than baseline Whisper on any reviewed batch of ≥ 20 Japanese-dominant clips, or $> 25\%$ of clips regressed.	Pause the pilot; inspect LID segmentations; revisit gates or retrain; resume only after a clean reevaluation.	Project owner (C. Bradshaw)
False-switch rate	Reviewers flag nonsense English insertions or sub-0.3 s switch artifacts in $> 10\%$ of a batch's clips.	Tighten the span-duration gate; prototype the third boundary class; reevaluate before restoring defaults.	Project owner
Coverage of new voices and styles	A new accent or speaking style shows repeated failures in correction logs.	Collect targeted consented recordings; add them to a held-out evaluation set before claiming coverage.	Project owner
Licensing exposure	Any proposal for commercial use, public release, or redistribution of models or audio.	Block until XTTS is replaced, contributor data is dropped or re-consented, and a rights review is complete.	Project owner

The project owner reviews correction logs at least monthly while the pilot is active and whenever a trigger fires, owns model and data versioning, and has authority to continue, restrict, retrain, roll back, or suspend the system. Beta testers own their individual transcripts; they do not inherit system-level monitoring responsibility.

Reevaluation And Update Triggers

Reevaluation is required before:

- enabling English-dominant transcription or changing the frame class list;
- changing splice-gate thresholds, the coalescing rule, or LID checkpoint;
- swapping the Whisper backbone (e.g., to `small` or `large-v3`);
- extending any claim to spontaneous conversation;
- any public or commercial release, which additionally requires the licensing actions above; or
- retraining after data regeneration, given the overwrite incident.

Every reevaluation must report speaker-disjoint frame metrics plus end-to-end WER/CER against baseline Whisper, split by dominant language, on both synthetic held-out speakers and consented real voices, with the checkpoint and manifest versioned so the prior model can be restored.

Most Important Constraints

The most important constraint is organizational, not architectural: every transcript remains a human-reviewed draft. The second is the rights chain — XTTS v2's non-commercial license, Common Voice reference clips, and informal contributor consent together cap deployment at research, personal use, and the consent-clean pilot. Operationally, training is a single 6h51m run on one Mac with the GPU pinned at 100% ($\approx 85^\circ\text{C}$ average); the Slurm cluster path failed silently and remains unresolved; and the project has one maintainer. A deployment with real users would need a reproducible training environment before it could be responsibly maintained.

Next Step Before Broader Use

Run the Japanese-dominant pilot on f1-974: first rerun the Table 3 evaluation with that checkpoint to confirm parity; recruit a small tester group under written terms; log corrections and refusals; build a consented real-voice held-out set; run the `whisper-large-v3` baseline comparison; and prototype the third boundary class for the

English-dominant problem. Broader use should be reconsidered only if that evidence supports it and the review and monitoring responsibilities can be maintained.

Evidence Appendix

Decision Trace

Table 6: Trace from report claims to evidence.

Report claim	Primary evidence
Frame metrics 0.9760 / 0.9743 on speaker-disjoint held-out data.	<code>models/lid-wav2vec2-f1-98/training_checkpoint.json</code> (saved 2026-07-07) and <code>models/lid-wav2vec2-f1-974/training_checkpoint.json</code> (saved 2026-07-05).
End-to-end WER/CER results (Table 3).	<code>just eval</code> runs, 2026-08-01: faster-whisper base int8, splice defaults, checkpoint f1-98, per-prefix slices.
Held-out split contains only synthetic speakers.	Replication of <code>split_examples()</code> (<code>cmd/train/dataset.py</code> , seed 13), 2026-08-01: 25 synthetic test speakers; all four manual speakers in train.
Dataset composition and label pipeline.	<code>data/data-pairs.csv</code> (31,634 rows); <code>docs/how-it-works.html</code> .
Reference-clip quality pivot; early checkpoint context.	Commits <code>cee6aac2c</code> and <code>40a92467d</code> (June 2026); author interview, 2026-08-01.
Manual-data overwrite and restoration.	Commits <code>42f587749</code> (restore, 2026-07-06) and <code>11073a64a</code> (guard); author interview.
Contributor system design and consent scope.	<code>docs/remote-recording-plan.md</code> ; author interview.
Training environment and cluster failure.	Author interview: Mac (MPS), 6h51m, GPU $\approx 85^\circ\text{C}$; <code>run.s</code> <code>slurm</code> job started but produced no output.
Splice gates and fail-toward-baseline behavior.	<code>cmd/eval/main.py</code> defaults and <code>--self-check</code> assertions.

Additional Evidence Links

- `docs/how-it-works.html` — full pipeline walkthrough: sentence factory, audio factory, contributor systems, training decisions, and the splice strategy.
- `docs/remote-recording-plan.md` — contributor recording architecture, locked design decisions, and phase-by-phase verification.
- `models/*/training_checkpoint.json` — per-checkpoint metrics and provenance for the two candidate models.
- **Weights & Biases logs** — training curves for all runs, including the June 24 40-epoch run behind f1-823.
- `cmd/eval/main.py --self-check` — executable assertions documenting coalescing, the midpoint-inside rule, and each splice gate.

Checkpoint Provenance And The Manual-Data Incident

Between July 4 and July 6, 2026, a synthetic-audio regeneration overwrote the 75 manual recordings with synthetic audio for the same sentences. Checkpoint f1-974 (saved July 5) was trained inside that window and therefore contains no contributor audio; the manual data was restored on July 6 (commit 42f587749) and f1-98 (saved July 7) trained on the full dataset. This report cites f1-98's 0.9760 macro-F1 as the current quality lead and designates f1-974 (0.9743) as the only checkpoint eligible for the beta pilot, since no consent question attaches to its training data. The 0.0017 quality difference does not change the recommendation; the consent difference does.